Atoms as Springs — Why Materials Stretch

Solid Mechanics · Lesson 1 of 5 · browser-based

1·Bond | 2·Spring | 3·Real spring | 4·Energy well

each tab adds exactly one idea

1 LEARN THE MATHEMATICS

build the model

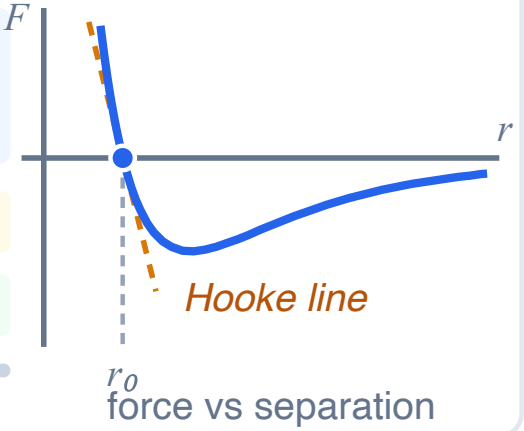
Background: two atoms, one tug-of-war

$$F = -k\,x,\ x = r - r_0$$

$$U(r) = 4\varepsilon[(\sigma/r)^{12} - (\sigma/r)^{6}]$$

worked example (copper)

key takeaways



force vs separation

2 DRIVE THE SIMULATION

test the model

scenario: Squeezed | Natural | **Stretched** | Near break

Controls

Separation 1.35

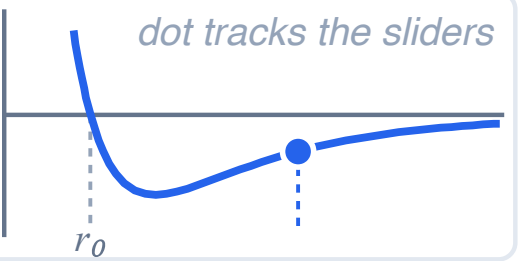
Bond strength 1.0

force F	−1.1 ε/σ
stiff. k	57 ε/σ²
energy U	−0.30 ε



atoms pull back toward r_0

dot tracks the sliders



✓ **self-check — automatic verdict**

"pull-back weakening — the bond is near fracture"

build the model → test it → revise

self-contained HTML — no install, runs on any device